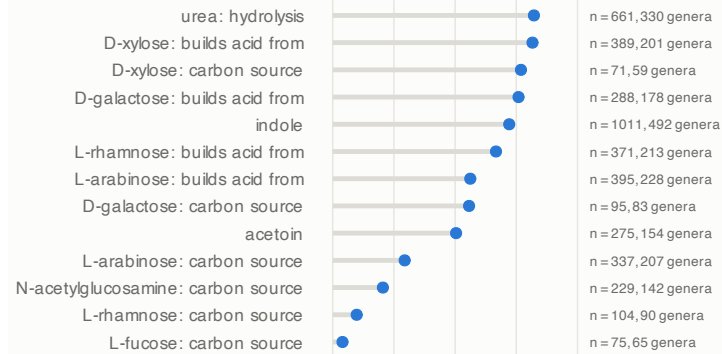


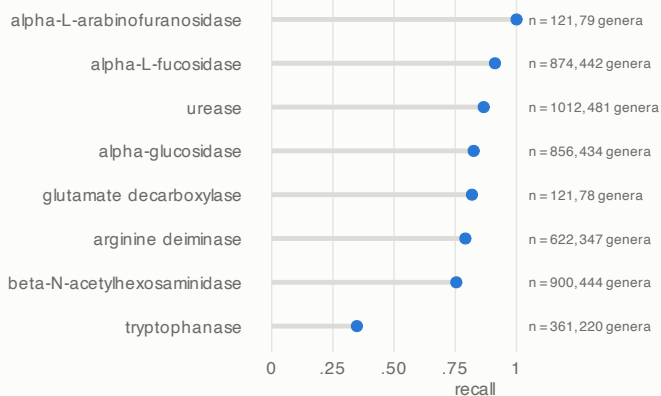
a Recall against observed phenotype

one row per curated mapping; nothing pooled

GIFT layer: substrate use or product formation

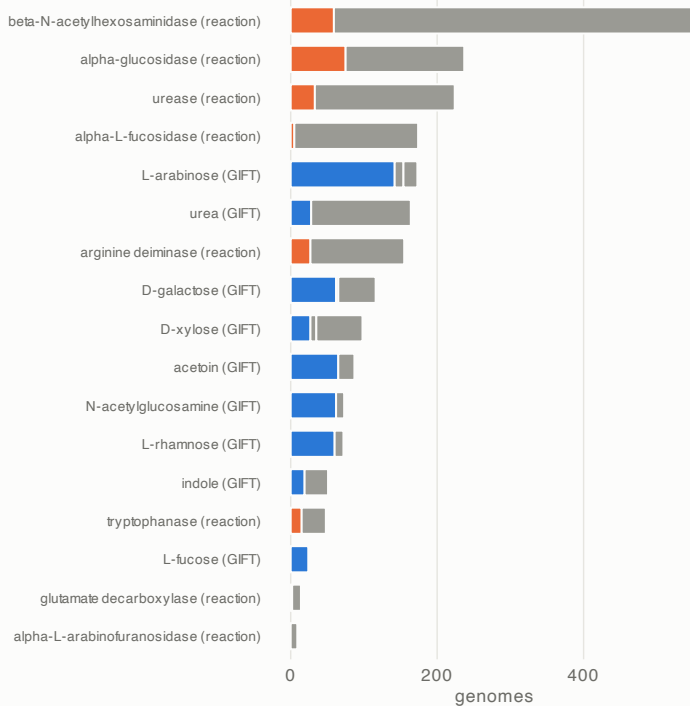


reaction layer: EC-resolved enzyme assay



c Every disagreement, classified

grey is permitted under invariant 15, never an error



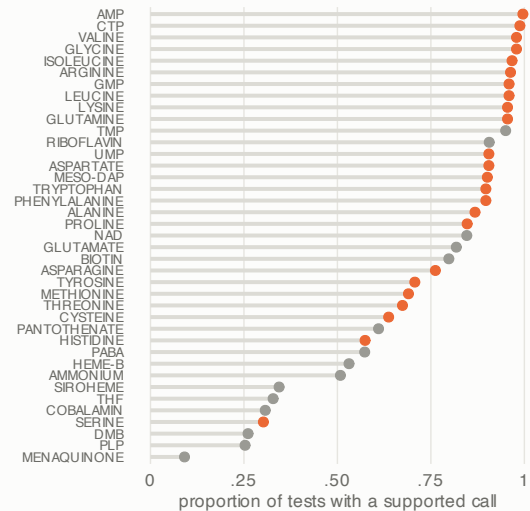
failure: no evidence at all

failure: incomplete route, step named

permitted: encoded, not observed

b Anabolic half: defined media

5803 tests on the essential classes; one point per nu

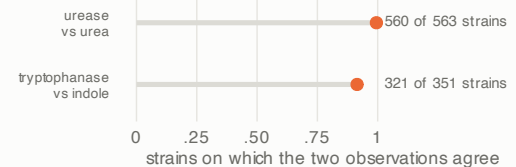


biomass-essential: the premise holds

not biomass-essential: premise fails

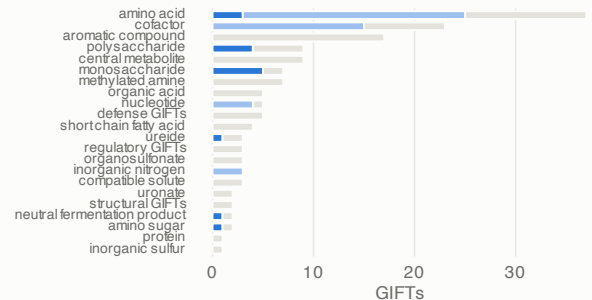
d The reference against itself

two independent assays of one capability



e Testable at all: 15 of 153 GIFTs

the unstable majority is named, not a remainder



individually testable

only as a bounded-frame aggregate

no phenotype reference at all

f The same genomes, three annotations

complete calls over 398 genomes

